

EE115 Linear Circuit Analysis (All sections)
Mid Term - I (FALL 2015)

Maximum Marks: 50

Instructor: Dr. Saima Zafar & Ms. Beenish Fatima

Total Time: 1 hr

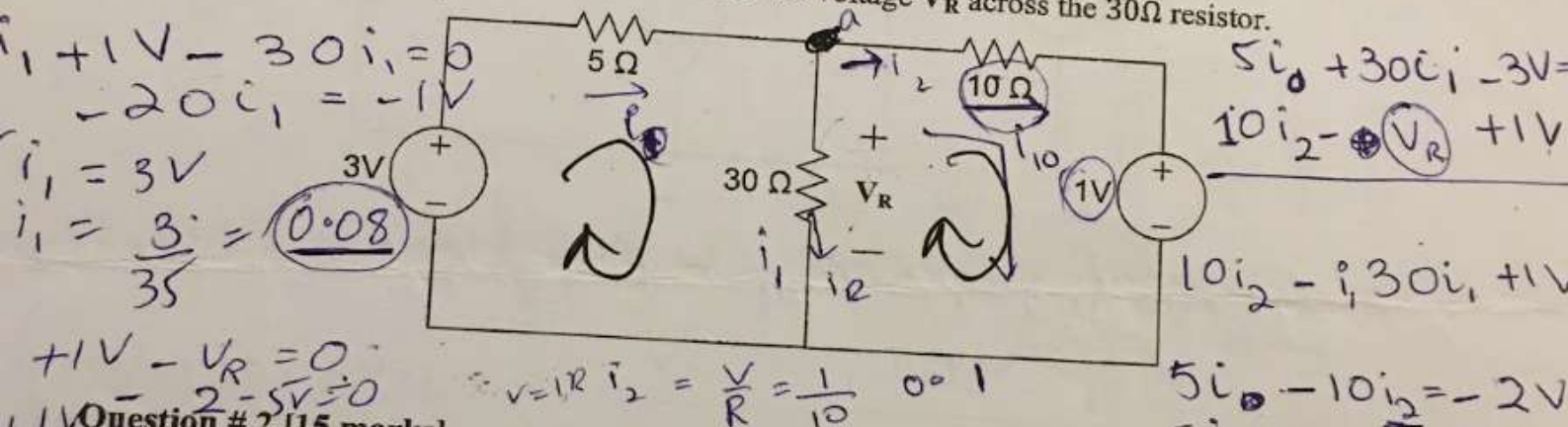
Dated: 15th September, 2015

INSTRUCTIONS: Carefully read the paper and attempt each question (only once) by using the technique mentioned in it. Show all calculations, direct answers are not acceptable. Make assumptions if you think some data is missing, but clearly mention your assumption. Your answers should be approximated up to two decimal places. Attempt all parts of a question together.

Question # 1 [15 marks]

Class Learning Objective: An ability to apply Ohm's Law and Kirchhoff's laws to analyze simple resistive circuits.

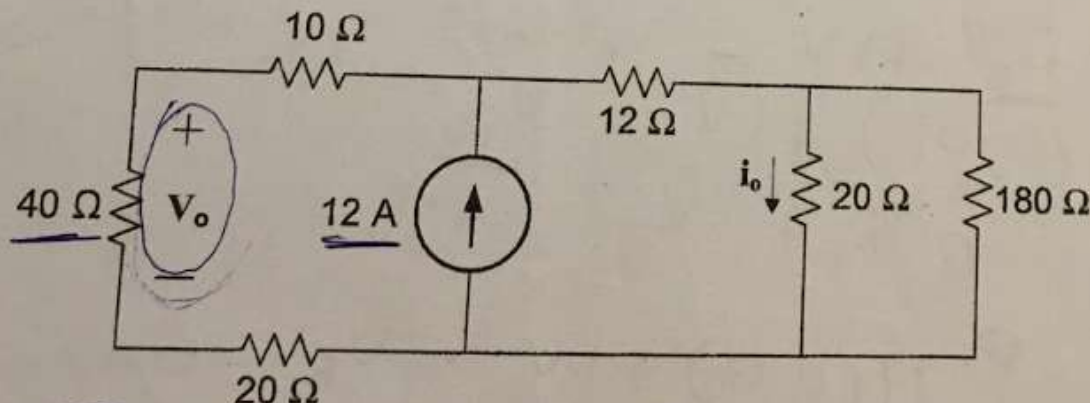
Use only **KCL**, **KVL** and **Ohm's Law** to find the voltage V_R across the 30Ω resistor.



Question # 2 [15 marks]

Class Learning Objective: An ability to simplify series, parallel circuits and apply CDR, VDR to compute unknown currents and voltages.

For the circuit shown in the figure below, use **current divider rule**, **voltage divider rule** and **Ohm's Law** only to calculate v_o , i_o and the **power** developed by the 12A current source.



$$5i_1 + 30i_2 - 3V = 0$$

$$10i_2 - V_R + 1V = 0$$

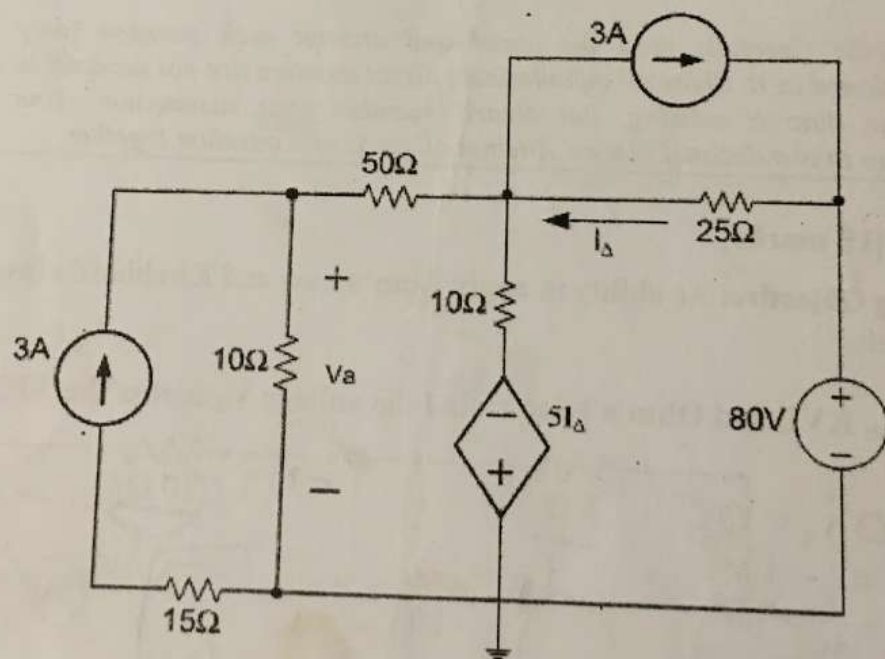
$$-i_1 + i_R + i_2 = 0$$

Question # 3 [16 + 4 = 20 marks]

Class Learning Objective: An ability to use node-voltage method to solve a circuit containing independent and/or dependent sources.

For the circuit given below, find the following using **Node Voltage** method:

- a) Voltage V_a
- b) Power associated with the 25Ω resistor.



$$(5\Omega)(i_o) + 30i_1 - 3V = 0$$